

The National Higher School of Artificial Intelligence
Introduction to AI Course Project
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**Algiers in 24 Hours:
A Time-Constrained Orienteering Approach
to Tourist Route Optimisation**

Full Name Group

Abstract

Provide a brief overview of the project to introduce the reader to your report.

1. Introduction

A brief introduction to the project problem you are addressing, contextualising the problem, motivating it, and explaining its importance.

2. State of the Art

3. Dataset Building

3.1 Data Collection

We created a dataset of 58 tourist landmarks in Algiers, classified into five landmark categories: historical, attractions, traditional and artistic, shopping, and religious sites.

Each entry in the dataset has GPS coordinates, the time the site is open to visitors, an estimate for how long it takes to visit, and a user rating from either Google Maps or TripAdvisor. Public sources were used to check the hours of operation, although much of the data was obtained from Google Maps directly. Table 1 presents the attributes, attribute types, and methods used to obtain the data.

Table 1: Dataset attribute descriptions.

Attribute	Type	Source	Description
Name	Text	Manually	Full official name of the landmark
Category	Text	Manually	Thematic group assigned to one of five classes
Latitude	Float	Google Maps	WGS84 decimal degrees, manually verified
Longitude	Float	Google Maps	WGS84 decimal degrees, manually verified
Opening hours	Text	Official listing	Structured time-window per day (e.g. 09:00–19:00)
Visit duration	Integer	Estimation	Expected time on-site in minutes
Google rating	Float	Google Maps	Aggregated user score on a 1–5 scale
TripAdvisor rating	Float	TripAdvisor	Aggregated user score on a 1–5 scale (missing for 18 sites)
Interest Score	Float	Computed	Google + TripAdvisor, or Google $\times$ 2 when TripAdvisor is missing, mapped to 1–10

3.2 Preprocessing

- **Coordinate verification:** All GPS coordinates were manually cross-checked using Google Maps satellite view to avoid positioning errors.
- **Interest score calculation:** An overall Interest Score was calculated on a 1–10 scale. When both ratings are available, the score is defined as Google + TripAdvisor; when only a Google rating exists (18 landmarks, 31%), the score is set to Google $\times$ 2, maintaining scale consistency.
- **Category assignment:** Each landmark was classified into one of five categories: historical, attraction, tradition & art, religious, and shopping. This enables interest-based filtering in the Day Planner application and improves the tourist experience.
- **Opening-hours standardisation:** Time windows used by the optimiser were converted into structured formats (e.g. 09:00–17:00). For venues with split opening hours (for example, 11:00–12:30 and 15:00–17:30), multiple time windows were defined for each day.

3.3 Descriptive Statistics and Characteristics of the Dataset

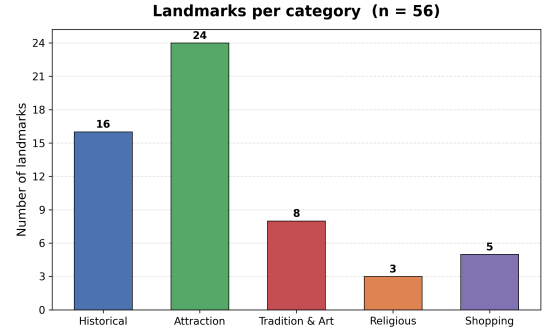
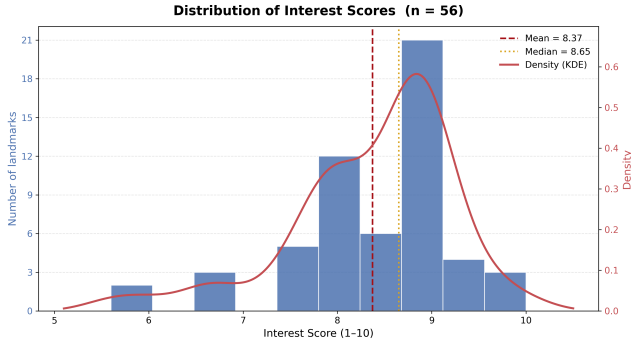
The dataset contains 58 unique landmarks divided into five categories. Numerical descriptive statistics are shown in Table 2, while the descriptive visualisations are shown in Figure 1 and Figure 2. Because two landmarks were marked as continuously closed, the visualisations are based only on the remaining 56 landmarks that are available to the planner.

The numerical summary is based on variables that directly influence recommendation and route optimisation: the calculated Interest Score, the two external rating sources, and the estimated visit duration. These statistics are used to measure the quality of the selected landmarks and the practicality of including them in daily itineraries.

Table 2: Descriptive statistics for numerical attributes.

Metric	Interest score	Google	TripAdvisor	Duration (h)
<i>n</i>	58	58	40	57
Mean	8.376	4.266	4.123	1.544
Median	8.600	4.400	4.200	1.500
Std dev	0.853	0.412	0.488	0.836
Min	5.600	3.000	2.000	0.500
Max	10.000	5.000	4.800	3.000

The Interest Scores range from 5.6 to 10.0 with an average of 8.38. This means that the data set is generally of high quality with good reviews. The Google and TripAdvisor ratings are highly correlated (averages of 4.27 and 4.12 respectively) which validates the summation-based scoring formula. The average visit duration of 1.54 h per landmark is directly fed to the optimiser when constructing feasible itineraries within a fixed time budget.

**Figure 1:** Distribution of Interest Scores with KDE overlay. Dashed = mean; dotted = median.**Figure 2:** Landmark count per category for the 56 available landmarks.

4. Problem Solving Techniques

5. Application Design

6. Results and Analysis

7. Discussion

The experimental results reveal a nuanced performance landscape in which solution quality, computational cost, and constraint satisfaction trade off in ways that both confirm and extend the findings reported in the orienteering literature.

7.1. Interpretation of Results

The most striking result is the parity between GRASP and Tabu Search at the highest observed tour quality of 98.7, both visiting 11 landmarks within the eight-hour budget, yet with a wall-clock time ratio of approximately 27:1 in favour of Tabu Search (0.37 seconds versus 10.18 seconds). This finding is consistent with the comparative analysis of **Liang2002**<empty citation>, who observed that Tabu Search and population-based heuristics achieve comparable solution quality on orienteering benchmarks, and with **Tang2005**<empty citation>, who demonstrated that adaptive memory mechanisms in Tabu Search enable rapid convergence through structured neighbourhood pruning. The successive-filtering candidate strategy implemented in our Tabu Search — ranking landmarks by induced travel cost relative to interest score — appears to be the principal factor responsible for its computational efficiency, effectively reducing the neighbourhood evaluation burden without sacrificing solution quality.

Simulated Annealing with the Cauchy acceptance criterion (96.9, 11 landmarks) consistently outperforms the Boltzmann variant (95.3, 11 landmarks), a result attributable to the heavier tails of the Cauchy distribution, which sustain a higher probability of accepting deteriorating moves at intermediate temperatures and thereby reduce premature convergence. This echoes the theoretical expectations discussed by **Gunawan2016**<empty citation>, who note that acceptance-function design is a primary differentiator of Simulated Annealing variants on constrained routing problems.

The Genetic Algorithm (68.3–90.0 across configurations) exhibits greater variance than single-solution methods, which is consistent with the sensitivity of population-based approaches to crossover operator design on time-windowed instances. The order crossover and tailored crossover implementations produced different fitness landscapes, and the relatively modest performance of the penalty fitness function relative to the feasibility-oriented variant suggests that soft-constraint handling significantly affects search trajectory for this problem class.

The most academically interesting result is the underperformance of the CPLEX exact solver (61.8–88.2, 7–10 landmarks) relative to GRASP and Tabu Search. Three explanations are plausible. First, CPLEX required 30.2 seconds to terminate, suggesting that the branch-and-bound search tree was not fully explored within the available computation budget, and the best integer solution found at termination is consequently a feasible but sub-optimal incumbent. Second, the Mixed Integer Linear Programming formulation introduces symmetry constraints and big- M linearisations for time-window propagation that tighten the feasible region relative to the heuristic representations. Third, the haversine-based travel matrix, while consistent across all solvers, produces a non-integer cost structure that may slow LP relaxation bounds. **Righini2009**<empty citation> observed similar behaviour and addressed it through decremental state-space relaxation; incorporating such techniques would likely close the gap between CPLEX and the leading metaheuristics.

7.2. Limitations

Several limitations constrain the generalisability of these results. First, travel times are computed from straight-line haversine distances at a fixed average urban speed of 25 km/h. Real road distances in Algiers, which the backend routing service retrieves from MAPBOX API, are systematically longer and temporally variable due to traffic. The algorithms were therefore optimising a surrogate objective that diverges from the true tourist experience. Second, landmark interest scores were assigned through a weighted combination of public ratings and expert judgement, a process that introduces subjectivity and limits reproducibility. Third, the dataset is confined to 56 landmarks in the Algiers wilaya, a scale at which all

tested algorithms remain tractable; results may not extrapolate to larger metropolitan datasets. Finally, the time-window representation assumes deterministic opening hours, whereas real tourist sites may close unexpectedly or have dynamic crowd-dependent effective visit durations.

7.3. Future Work

Several directions offer meaningful improvements. Within the classical metaheuristic paradigm, Adaptive Large Neighbourhood Search (ALNS) represents the current state of the art for time-constrained orienteering (Gunawan2016), with its self-tuning destroy-and-repair operators addressing the principal weakness of both GRASP (construction diversity) and Tabu Search (neighbourhood breadth). Iterated Local Search with the MaxShift data structure of Vansteenwegen2009<empty citation> would provide an efficient feasibility oracle that reduces the per-iteration cost of neighbourhood evaluation from $O(n)$ to $O(1)$.

At the frontier of the field, deep reinforcement learning (DRL) approaches have demonstrated near-optimal performance on routing problems of this scale. Kool2019<empty citation> showed that an Attention Model trained with the REINFORCE policy gradient algorithm learns generalisable construction heuristics for TSP and related problems; Gama2020<empty citation> extended this framework directly to the OPTW. Hybrid DRL-ALNS architectures, in which a learned policy selects neighbourhood operators dynamically (Reijnen2024), represent the most promising direction for achieving both solution quality and sub-second inference times. A further extension of direct practical relevance is the Stochastic Orienteering Problem, in which travel times are drawn from a distribution reflecting real traffic conditions; this formulation is underexplored in the literature (Vansteenwegen2011) and maps naturally onto the Algiers urban context. Finally, extending the single-day formulation to the Team Orienteering Problem with Time Windows (Gunawan2016) would enable multi-day, multi-guide itinerary planning, substantially increasing the practical utility of the system.

8. Conclusion

This paper presented *Algiers in 24 Hours*, a tourist route optimisation system grounded in the Time-Constrained Orienteering Problem with Time Windows. Six algorithmic approaches spanning the full spectrum from deterministic construction to exact branch-and-bound were implemented, integrated into a common model layer, and evaluated on a curated dataset of 56 Algiers landmarks with verified GPS coordinates, opening hours, and interest scores.

The principal finding is that Tabu Search with Strategic Oscillation achieves solution quality equivalent to GRASP (98.7 interest score, 11 landmarks) at a fraction of the computational cost (0.37 versus 10.18 seconds), establishing it as the recommended solver for real-time deployment contexts. Simulated Annealing with the Cauchy acceptance criterion provides a competitive alternative (96.9) with moderate runtime. The underperformance of CPLEX relative to leading metaheuristics underscores the known intractability of the OPTW for branch-and-bound approaches under realistic time budgets, and motivates the design of hybrid exact-heuristic methods as a direction for future investigation.

Beyond algorithmic contribution, this work produces a reusable, open-source problem model class for the Algerian tourist orienteering context — a domain absent from existing benchmark libraries — and a full-stack web application through which algorithm behaviour can be explored interactively by non-expert users. The gap between the best heuristic score and the CPLEX incumbent across all experiments defines a concrete quality target for future work. Closing that gap through Adaptive Large Neighbourhood Search or learned construction heuristics, while extending the formulation to multi-day and stochastic settings, constitutes the principal agenda for subsequent research.

A. Screenshots of the System

B. Who Did What

Table 3: Team member contributions.

Member	Primary Contributions
Raouf Ould Ali	Integrate Frontend, Implement Simulated Annealing, Implement CPLEX Solution, Setup GitHub Repository, Review Performance Analysis, Review Grasp Search, Write Application Design (Report), Review Dataset Building (Report), Review Conclusion (Report), Write References (Report).
Mohamed Zakaria Razam	Implement Frontend, Implement Greedy Search Algorithm, Write Code Documentation, Review Simulated Annealing, Review CPLEX Solution, Write Dataset Building (Report), Review Application Design (Report), Review Discussion (Report), Write Appendix B (Report).
Ahmed Adnane Meddah	Review Frontend, Implement Performance Analysis & Visualization, Implement Unit Testing, Review Greedy Search, Write Introduction (Report), Review Appendix A (Report), Review Appendix B (Report), Review Jupyter Notebook.
Mohamed Charaf Eddine Deghboudj	Backend Development, Implement Problem Model Class, Implement Grasp Search, Review Genetic Algorithm, Review Tabu Search, Review State of the Art (Report), Review Results and Analysis (Report), Write Discussion (Report), Write Conclusion (Report).
Yacine Mouloud	Backend Development, Implement Problem Model Class, Implement Tabu Search, Implement Unit Testing, Review Abstract (Report), Review Problem Solving Techniques (Report), Write Results and Analysis (Report), Write Appendix A (Report).
Aboubakr Bendahmane	Implement Genetic Algorithm, Setup L ^A T _E X Template, Review Problem Model, Review Unit Testing, Write Abstract (Report), Write State of the Art (Report), Write Problem Solving Techniques (Report), Review Introduction (Report), Review References (Report).